

Revised Generation Interconnection System Impact Study Report

For Queue Project Q11-0063

Boulder Basin 138 kV
November 16, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0063. The Interconnection Customer proposes a 1130.5 MW natural gas combined cycle facility interconnecting at the Boulder Basin 138 kV substation in the Maryland area. The request was received on August 07, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 56 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$73,805,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of BASELINE - BOULDER BASIN 138kV, the ISHUA SWITCH TOWER - BOULDER BASIN 138kV L loads to 582.3% of its 189 MVA rating. Project DFAX: 100.0%.

BOULDER BAIN - BASELINE 138kV: 527.1% loading (208 MVA) following the outage of the ISHUA SWITCH TOWER - BOULDER BASIN 138kV L; DFAX = 100.0%.

For the loss of ISHUA SWITCH TOWER - BOULDER BASIN 138kV, the BASELINE - BECO 138kV loads to 480.3% of its 223 MVA rating. Project DFAX: 100.0%.

For the loss of BOULDER BASIN - ISHUA SWITCH TOWER 138kV, the BECO - STERLING PARK 138kV loads to 453.8% of its 230 MVA rating. Project DFAX: 100.0%.

For the loss of BOULDER BASIN - BASELINE 138kV, the RADFORDS RUN WIND FARM - ST FRANCOIS SUBSTATION 138kV loads to 346.2% of its 167 MVA rating. Project DFAX: 54.4%.

The HOLLYDALE - ST FRANCOIS 138kV is overloaded to 317.8% of its 186 MVA emergency rating for the loss of the BOULDER BASIN - BASELINE 138kV. The project contributes 54.4% DFAX (614.9 MW) to this overload.

The HOLLYDALE - ISHUA SWITCH TOWER 138kV is overloaded to 300.7% of its 207 MVA emergency rating for the loss of the BOULDER BASIN - BASELINE 138kV line. The project contributes 54.4% DFAX (614.9 MW) to this overload.

The DAVIS DRIVE - STERLING PARK 138kV is overloaded to 236.9% of its 311 MVA emergency rating for the loss of the ISHUA SWITCH TOWER - BOULDER BASIN 138kV. The project contributes 76.6% DFAX (865.7 MW) to this overload.

The ISHUA SWITCH TOWER - LON WRIGHT 138kV is overloaded to 228.3% of its 217 MVA emergency rating for the loss of the ISHUA SWITCH TOWER - HOLLYDALE 138kV. The project contributes 46.3% DFAX (523.6 MW) to this overload.

For the loss of ISHUA SWITCH TOWER - HOLLYDALE 138kV line, the CHEATHAM - LON WRIGHT 138kV line loads to 221.5% of its 220 MVA rating. Project DFAX: 46.3%.

The DAVIS DRIVE - DRANESVILLE 138kV L is overloaded to 174.0% of its 397 MVA emergency rating for the loss of the BOULDER BASIN - ISHUA SWITCH TOWER 138kV L. The project contributes 76.6% DFAX (865.7 MW) to

this overload.

DARIN - COLESVILLE SUBSTATION 138kV Line: 165.0% loading (167 MVA) following the outage of the ADRIAN ENERGY ASSOCIATES LLC 345-138 kV Transformer; DFAX = 7.2%.

For the loss of ISHUA SWITCH TOWER - LON WRIGHT 138kV L, the RADFORDS RUN WIND FARM - HADDAM NECK 138kV loads to 158.0% of its 235 MVA rating. Project DFAX: 21.3%.

The HADDAM NECK - LEADVILLE 138kV Line is overloaded to 151.8% of its 227 MVA emergency rating for the loss of the LON WRIGHT - ISHUA SWITCH TOWER 138kV. The project contributes 21.3% DFAX (240.5 MW) to this overload.

For the loss of ISHUA SWITCH TOWER - LON WRIGHT 138kV, the BELLS MILL SUBSTATION - LONGWOOD SUBSTATION 138kV loads to 150.0% of its 172 MVA rating. Project DFAX: 21.3%.

LEADVILLE - POTOMAC 138kV: 146.1% loading (218 MVA) following the outage of the LON WRIGHT - ISHUA SWITCH TOWER 138kV; DFAX = 21.3%.

The LONGWOOD SUBSTATION - POTOMAC 138kV is overloaded to 143.2% of its 197 MVA emergency rating for the loss of the ISHUA SWITCH TOWER SUBSTATION - LON WRIGHT SUBSTATION 138kV line. The project contributes 21.3% DFAX (240.5 MW) to this overload.

LEESBURG - STERLING PARK 138kV: 143.0% loading (188 MVA) following the outage of the BOULDER BASIN - ISHUA SWITCH TOWER 138kV; DFAX = 23.4%.

For the loss of ADRIAN ENERGY ASSOCIATES LLC 345/138kV Transformer Bank, the SUNBRIGHT - COLESVILLE 138kV loads to 135.5% of its 207 MVA rating. Project DFAX: 7.1%.

CANOVANAS PUEBLO 2401 - LOW GAP 138kV: 127.9% loading (166 MVA) following the outage of the ISHUA SWITCH TOWER - HOLLYDALE 138kV; DFAX = 20.9%.

For the loss of ISHUA SWITCH TOWER - HOLLYDALE 138kV, the CHEATHAM - CANOVANAS PUEBLO 2401 138kV loads to 123.2% of its 204 MVA rating. Project DFAX: 20.9%.

The MCCORMICK - FOSTER DAM 138kV is overloaded to 118.9% of its 246 MVA emergency rating for the loss of the HELIJET 345/138kV Auto. The project contributes 5.4% DFAX (61.5 MW) to this overload.

For the loss of CARDEROCK - ELK GARDEN 138kV line, the HELIJET - ELK GARDEN 138kV loads to 112.1% of its 219 MVA rating. Project DFAX: 6.5%.

The FOSTER DAM - ELK GARDEN 138kV line is overloaded to 111.0% of its 234 MVA emergency rating for the loss of the HELIJET 345/138 kV Transformer Bank. The project contributes 5.4% DFAX (61.5 MW) to this overload.

WILKENS - WHEELABRATOR BALTIMORE REFUSE 138kV: 108.2% loading (252 MVA) following the outage of the TABER - DIXLETA 345kV line; DFAX = 5.3%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ISHUA SWITCH TOWER - BOULDER BASIN 138kV L	Rebuild line with higher-capacity conductor	\$2,259,000	100% (\$2,259,000)
BOULDER BASIN - BASELINE 138kV	Rebuild line with higher-capacity conductor	\$4,326,000	100% (\$4,326,000)
BASELINE - BECO 138kV L	Rebuild line with higher-capacity conductor	\$3,292,000	100% (\$3,292,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
STERLING PARK - BECO 138kV	Rebuild line with higher-capacity conductor	\$2,139,000	100% (\$2,139,000)
RADFORDS RUN WIND FARM - ST FRANCOIS SUBSTATION 138kV	Rebuild line with higher-capacity conductor	\$3,240,000	100% (\$3,240,000)
HOLLYDALE - ST FRANCOIS 138kV	Rebuild line with higher-capacity conductor	\$1,191,000	100% (\$1,191,000)
HOLLYDALE - ISHUA WITCH TOWER 138kV	Rebuild line with higher-capacity conductor	\$2,504,000	100% (\$2,504,000)
DAVIS DRIVE - STERLING PARK 138kV	Rebuild line with higher-capacity conductor	\$1,986,000	100% (\$1,986,000)
LON WRIGHT - ISHUA SWITCH TOWER 138kV	Rebuild line with higher-capacity conductor	\$3,597,000	100% (\$3,597,000)
CHEATHAM SUBSTATION - LON WRIGHT 138kV	Rebuild line with higher-capacity conductor	\$3,364,000	100% (\$3,364,000)
DAVIS DRIVE - DRANESVILLE 138kV L	Rebuild line with higher-capacity conductor	\$2,325,000	100% (\$2,325,000)
COLESVILLE - DAIEN SUBSTATION 138kV L	Rebuild line with higher-capacity conductor	\$919,000	100% (\$919,000)
RADFORDS RUN WIND FARM - HADDAM NECK 138kV	Rebuild line with higher-capacity conductor	\$1,508,000	100% (\$1,508,000)
HADDAM NECK - LEADVILLE 138kV Line	Rebuild line with higher-capacity conductor	\$3,465,000	100% (\$3,465,000)
BELLS MILL SUBSTATION - LONGWOOD SUBSTATION 138kV	Rebuild line with higher-capacity conductor	\$2,399,000	100% (\$2,399,000)
POTOMAC - LEADVILE SUBSTATION 138kV Line	Rebuild line with higher-capacity conductor	\$2,323,000	100% (\$2,323,000)
LONGWOOD SUBSTATION - POTOMAC 138kV	Rebuild line with higher-capacity conductor	\$2,714,000	100% (\$2,714,000)
LEESBURG - STERLING PARK 138kV	Rebuild line with higher-capacity conductor	\$10,582,000	100% (\$10,582,000)
COLESVILLE - SUNBRIGHT SUBSTATION 138kV	Rebuild line with higher-capacity conductor	\$1,284,000	76% (\$975,000)
CANOVANAS PUEBLO 2401 - LOW GAP 138kV	Rebuild line with higher-capacity conductor	\$7,639,000	100% (\$7,639,000)
CANOVAAS PUEBLO 2401 - CHEATHAM 138kV	Rebuild line with higher-capacity conductor	\$3,828,000	100% (\$3,828,000)
FOSTER DAM - MCCORMICK 138kV	Rebuild line with higher-capacity conductor	\$1,093,000	100% (\$1,093,000)
HELIJET - ELK GARDEN 138kV	Rebuild line with higher-capacity conductor	\$1,601,000	100% (\$1,601,000)
FOSTER DAM - ELK GARDEN 138kV line	Rebuild line with higher-capacity conductor	\$4,162,000	100% (\$4,162,000)
WILKENS - WHEELABRATOR BALTIMORE REFUSE 138kV L	Reconductor line	\$536,000	70% (\$374,000)
	Total allocated to project		\$73,805,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.